Beyond Linear Reasoning: Combining Chain-of-Thought with A* Search for Robust Multi-Strategy Inference

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Abstract

Chain-of-thought (CoT) prompting generates a single reasoning trajectory, while search-based methods like A* explore multiple partial traces before committing. Which approach should an LLM use for a given problem? We find that neither dominates: across StrategyQA, LogicQA, HotpotQA, GSM8K, and MATH, CoT and A* each solve substantial instance subsets that the other misses. To explain this complementarity, we develop a *Reasoning Topology Framework* built on 13 structural features computable from the input alone. A* advantage turns out to track a single derived quantity—branching coefficient—rather than dataset identity ($R^2 = 0.94$ across topology clusters). Building on this, we train a topology-aware router that sees no generated traces yet outperforms semantic-entropy and LLM-as-critic routers that inspect full reasoning chains, recovering 26.1% of the oracle gap. We trace this counterintuitive result to a *Fluency-Correctness Asymmetry*: in 38% of disagreement cases the more fluent trace is actually wrong, systematically misleading post-hoc judges. Sensitivity sweeps, compute accounting, and 14B-scale experiments confirm that the topology-dependent pattern is robust. A* is not a universal upgrade over CoT; it is a higher-cost strategy best deployed when problem structure rewards branching. Code: `https://anonymous.4open.science/r/astar-reasoning-EC39`

1 Introduction

Chain-of-thought (CoT) prompting (Wei et al., 2022) has become the standard way to elicit step-by-step reasoning from large language models (LLMs). By generating intermediate reasoning steps before a final answer, CoT often improves performance on arithmetic, commonsense, and symbolic tasks. However, its core generation process is still strictly *linear*: the model generates a single trace $s_1 \to s_2 \to \cdots \to s_T$, and later steps are conditioned on all earlier generated contents. This design creates three potential problems. First, an early mistake can propagate through the entire trace because the model has no built-in backtracking mechanism. Second, questions that require integrating several independent pieces of evidence can be difficult to solve with a single left-to-right narrative. Third, CoT has no explicit search objective that trades off trace length and alternative hypotheses.

These limitations motivate search-based reasoning. If the model can explore multiple partial traces, then it can reconsider early commitments, compare candidate branches, and prioritize promising reasoning states. In this paper, we study a budgeted A*-search procedure over natural-language reasoning traces. The language model generates successor steps, a step cost penalizes both trace length and low-confidence steps, and a heuristic prioritizes which partial trace to expand next.

But search is not free. Exploring multiple branches can also waste computation on dead-end decompositions or introduce noise on problems that a single well-chosen chain would solve directly. In our experiments, we find that A* occasionally *hurts* by overthinking simple questions (Section 3.5). The central question is therefore not "is search better?" but rather: *which structural properties of a reasoning problem predict whether branching will help?*

Concretely, we ask: (i) Does guided search dominate CoT, or do they trade wins? (ii) When they disagree, is the split random or systematic? (iii) Can we predict which method will win from structural features of the input alone—before generating any trace? (iv) Does such a pre-generation router actually outperform post-hoc adjudication that sees full traces?

Motivating example. On StrategyQA, linear CoT can pursue a locally plausible but ultimately irrelevant chain, while A* can surface the decisive evidence earlier by exploring alternative partial traces. For example (§Figure 1), on the question "Is shrimp scampi definitely free of plastic?", CoT misses the key fact of microplastic accumulation in seafood, whereas A* reaches that premise early and answers correctly. This illustrates the central hypothesis of the paper: CoT and search are complementary, and the real challenge is deciding when to use which.

This pattern persists at scale. Across all three benchmarks, CoT and A* solve different subsets of examples. On StrategyQA, A* outperforms CoT (74.80% vs. 64.52%), yet the oracle reaches 84.59%. On LogicQA, CoT is slightly stronger (45.78% vs. 44.85%), but the oracle still reaches 59.29%. On HotpotQA, A* again leads (80.10% vs. 76.80%), while the oracle reaches 94.20%. Together, these gaps show that the main opportunity is *strategy selection*: different questions favor different reasoning algorithms.

Why A* specifically? Our goal is not to champion A* as the best search algorithm for LLMs, but to use it as a controlled diagnostic. A* separates the key ingredients of guided search—a priority frontier, a path cost, and a heuristic estimate—without entangling them with evaluator design (as in ToT), rollout policies (as in MCTS), or fixed-width pruning (as in beam search). This makes it easier to isolate *when* the act of branching itself is responsible for gains.

Contributions. Our contributions: (1) We compare CoT and A* across five benchmarks (StrategyQA, LogicQA, HotpotQA, GSM8K, MATH) and three model scales (0.5B, 8B, 14B), finding that neither uniformly dominates. (2) We quantify instance-level complementarity: CoT and A* solve genuinely different subsets, and the oracle gap is large enough to make routing worthwhile. (3) We introduce 13 pre-generation topology features and show that a single derived quantity—branching coefficient—predicts A* advantage better than dataset identity ($R^2 = 0.94$). (4) We show that a topology-aware router outperforms post-hoc trace evaluation, and diagnose why through a Fluency-Correctness Asymmetry analysis.

2 Methodology

We first formalize CoT and search-based reasoning under a common notation and then describe the routing mechanisms that choose between them.

2.1 Notation and Problem Setup

Let $\mathcal{Q} = (q, C, \mathcal{O})$ denote a reasoning problem, where q is the question, C is an optional context passage, and $\mathcal{O} = \{o_1, \ldots, o_K\}$ is the candidate answer set for multiple-choice tasks or the output space for free-form tasks. A *reasoning trace* $\tau = (s_1, s_2, \ldots, s_T)$ is an ordered sequence of natural-language reasoning steps, and a prediction function $\phi(\tau) \in \mathcal{O}$ extracts the final answer from the trace. Let $\mathcal{M}_\theta$ denote a language model with parameters θ.

2.2 Chain-of-Thought (CoT) Reasoning

In CoT prompting, the model generates a single reasoning trace autoregressively:

$$\begin{aligned} \tau^{\text{CoT}} &= (s_1, s_2, \ldots, s_T) \\ s_t &\sim \mathcal{M}_\theta(\cdot \mid q, C, \mathcal{O}, s_{1:t-1}). \end{aligned} \quad (1)$$

Generation terminates when the model emits a terminal answer pattern, and the final prediction is $\hat{y}^{\text{CoT}} = \phi(\tau^{\text{CoT}})$. CoT explores exactly one element of the trace space with no mechanism for backtracking or branch comparison.

2.3 A* Search over Reasoning Traces

We therefore search over the space of reasoning traces using an A*-style priority rule, but with a finite node budget and a multi-goal voting procedure for tractability.

State space. The search space is a tree $\mathcal{G} = (\mathcal{S}, \mathcal{E})$ where each node $n \in \mathcal{S}$ represents a partial reasoning state:

$$n = (\tau_n, d_n), \qquad \tau_n = (s_1, \ldots, s_{d_n}). \quad (2)$$

Here, τ_n is the partial trace stored at node n and $d_n = |\tau_n|$ is its depth. The root node n_0 has $\tau_{n_0} = \emptyset$ and $d_{n_0} = 0$. An edge $(n, n') \in \mathcal{E}$ exists when $\tau_{n'} = \tau_n \oplus s$ for some reasoning step s.

Successor generation. Given a node n with state (τ_n, d_n), we ask the language model to propose N_c candidate next steps:

$$\{(s^{(j)}, c^{(j)})\}_{j=1}^{N_c} = \text{GS}(\mathcal{M}_\theta, q, C, \mathcal{O}, \tau_n, N_c). \quad (3)$$

Motivating Example *Query: "Is shrimp scampi definitely free of plastic?"* **Gold: FALSE**

✗ Chain-of-Thought *Linear generation*

Commits to a chronological schema, missing the biological premise.

S1. Shrimp are caught/farmed; no conclusive evidence of significant plastic.
S2. Processing does not involve contamination.
S3. Cooking does not introduce plastic.
S4. No evidence of ingredient contamination.
S5. Regulatory standards make contamination unlikely.
Ans. YES **(Incorrect)**

✓ A* Search *Heuristic-guided*

Prioritizes the key biological premise directly.

S1. Shrimp often contain microplastics in tissues.
S2. Preparation does not remove pre-existing microplastics.
Ans. FALSE **(Correct)**

Figure 1: Comparison of reasoning trajectories on a StrategyQA example. CoT follows a plausible but misdirected chronological schema, while A* prioritizes the key biological factmicroplastic presence in shrimpand reaches the correct conclusion with fewer, more relevant steps.

GS refers to the step-generation function. Each candidate consists of step text $s^{(j)}$ and a self-reported confidence $c^{(j)} \in [0.0, 1.0]$. The candidate induces a successor node n'_j with trace $\tau_{n'_j} = \tau_n \oplus s^{(j)}$ and depth $d_{n'_j} = d_n + 1$.

Goal condition. A node is a goal node if its last step contains a terminal answer pattern:

$$\textsc{IsGoal}(n) = \mathbb{1}\left[\exists p \in \mathcal{P} : p \subset s_{d_n}\right], \quad (4)$$

where $\mathcal{P}$ contains answer templates such as "The answer is [A/B/C/D]" for multiple-choice tasks.

Path cost. The cumulative cost of a node trades off reasoning length and uncertainty:

$$g(n) = \sum_{t=1}^{d_n} \left[1 + (1 - c_t)\right] = d_n + \sum_{t=1}^{d_n} (1 - c_t). \quad (5)$$

Each step contributes a base cost of 1, and lower-confidence steps incur an additional penalty. This makes shorter, higher-confidence traces cheaper than longer or more uncertain ones.

2.4 Heuristic Design

The heuristic estimates the remaining cost from a partial trace to a valid goal. We use a *critic-based* heuristic as the main experimental heuristic:

$$h(n) = 1 - \mathrm{score}(n), \quad (6)$$

where $\mathrm{score}(n) \in [0, 1]$ is a critic model's assessment of the partial reasoning state n for factual consistency and reasoning completeness. Higher scores indicate greater progress toward a correct conclusion, so $(1 - \mathrm{score}(n))$ estimates the remaining fraction of reasoning effort.

Approximate conservative guidance. Classical admissibility requires $h(n) \leq h^*(n)$ for all n. Since the critic is a learned model, we cannot guarantee this formally. Instead, we identify three sufficient conditions under which the heuristic behaves conservatively in practice: (i) *Monotone calibrated scoring*: if the critic score increases monotonically with true reasoning progress, then $1 - \mathrm{score}(n)$ is a valid upper bound on the fraction of remaining work. (ii) *Conservative estimation*: if the critic systematically underestimates progress, then $h(n)$ overestimates remaining cost, which is safe for search ordering (it delays expansion of promising nodes but never skips them). (iii) *Bounded overestimation*: even if the critic occasionally overestimates progress, as long as the overestimation is bounded by $\epsilon_{\max}$, the search remains approximately admissible with bounded suboptimality.

Empirical validation. We validate the heuristic on 500 annotated intermediate reasoning states (100 StrategyQA, 100 HotpotQA, 300 LogicQA). On average, the critic underestimates human-assessed progress (mean critic score: 0.43 vs. human assessment: 0.58; paired t-test $p < 0.001$), with an overestimation rate of only 12.3% and maximum observed overestimation $\epsilon_{\max} = 0.14$. This supports the interpretation that the heuristic operates in the conservative regime: it tends to underestimate how far along a reasoning state actually is, which biases search toward continuing exploration rather than premature termination.

An auxiliary depth-plus-coverage heuristic with a formal admissibility proof under explicit assumptions is provided in Appendix A for theoretical completeness. That construction applies to a simplified setting and is not used as the main experimental heuristic.

For the main experiments, we keep the search hyperparameters fixed across datasets to avoid over-tuning the search procedure itself. In particular, the reported results should be interpreted as evaluating a single budgeted A*-style configuration rather than an extensively tuned search system. Dataset wise A* search hyperparameter values are shown in Table 6.

Algorithm 1 summarizes the overall inference procedure.

2.5 Reasoning Topology Features

To characterize the structural demands of a reasoning problem *before* any model generation, we define 13 pre-generation features computed directly from the input text: question length, context length, context sentences, logical connectives, negation markers, comparatives, question marks, estimated hop count (dependency-based), contextual density (entities/sentence), distractors, reasoning depth, number of options, and a key derived feature—the *branching coefficient* bc = c_sent × hop/q_len—which captures how much branching structure the input affords. These features require no model inference and are used for both topology clustering (§3.3) and pre-generation routing (§3.4).

2.6 Routing Strategies

Given the complementarity between CoT and A*, we define routing functions $\mathcal{R} : \mathcal{Q} \to \{\text{CoT}, A^*\}$. We distinguish two families.

require generating traces before deciding. *Semantic entropy* (Kossen et al., 2024): routes high-entropy (under self-consistency) instances to A*. *LLM-as-critic*: on disagreements, a critic model (DeepSeek-R1-Distill-Qwen-14B, chosen for architectural distinctness from the LLaMA-3.1-8B backbone and local executability) compares both traces:

$$\text{verdict} = \mathcal{M}_{\text{critic}}(\text{Trace}_{\text{CoT}}, \text{Trace}_{A^*}, q, C, \mathcal{O}) \quad (7)$$

decide before any generation. *Random forest*: trained on hand-crafted features (question length, connectives, negation). *Topology router*: XGBoost on the 13 topology features, requiring no LLM inference at routing time. It is a diagnostic test of whether the CoT/A* boundary is predictable from problem structure.

3 Results

We evaluate on three benchmarks using two models: i) LLaMA-3.1-8B and ii) Qwen-0.5B and DeepSeek-R1-Distill-Qwen-14B as the critic. StrategyQA (Geva et al., 2021) tests multi-step commonsense reasoning with boolean answers. LogicQA (Liu et al., 2020) tests formal logical reasoning with four-way multiple choice. HotpotQA (Yang et al., 2018) tests multi-hop question answering with free-form answers. We report the exact accuracies from the current experimental pipeline, and the oracle corresponds to instance-wise selection of the correct answer between CoT and A*.

3.1 Quantitative Analysis

Main results. Table 1 shows that no single reasoning strategy dominates. A* is substantially stronger than CoT on StrategyQA (+10.28 pp) and HotpotQA (+3.30 pp), while CoT has a slight edge on LogicQA (+0.93 pp). Yet the oracle sits far above either strategy—improving by +9.79, +13.51, and +14.10 points respectively—confirming strong complementarity. The presence of substantial CoT-only and A*-only cells in Table 2 shows the methods implement genuinely different computations, motivating the topology analysis (Section 3.3) and routing (Section 3.4).

Router performance. Routing closes only part of the oracle gap. The feature-based random forest reaches 76.25% on StrategyQA (+1.45 over A*) and 48.20% on LogicQA (+2.42 over CoT). These gains are real but much smaller than the oracle margin. The LLM-as-critic router runs only

Dataset	Base methods on LLaMA-3.1-8B				Base methods on Qwen-0.5B				Best Router	Oracle
	CoT	A*	SC	ToT	CoT	A*	SC	ToT	RF	
StrategyQA	64.52	74.80	71.22	72.24	53.60	53.40	54.21	52.24	**76.25**	84.59
LogicQA	45.78	44.85	46.10	46.21	24.00	20.20	22.10	21.19	**48.20**	59.29
HotpotQA	76.80	80.10	78.21	78.45	8.40	30.00	12.21	22.17	**81.25**	94.20

Table 1: Main results across the three benchmarks. SC = Self-Consistency, ToT = Tree-of-Thought, RF = rule-based random forest router (evaluated on LLaMA-3.1-8B). The topology router (Table 4) further improves over RF on StrategyQA and LogicQA. Bold = best non-oracle; underlined = best single method. All router gains over CoT: McNemar's $p < 0.005$.

on the ∼25% of disputed cases but gains are modest, suggesting that adjudicating between traces is difficult—especially when A* traces are less fluent yet logically stronger (analyzed further in Section 3.4).

Dataset	Both correct	CoT only	A* only	Both wrong
StrategyQA (n=2290)	1361 (59.4%)	328 (14.3%)	248 (10.8%)	353 (15.4%)
LogicQA (n=651)	180 (27.6%)	118 (18.1%)	88 (13.5%)	265 (40.7%)
HotpotQA (n=500)	298 (59.6%)	86 (17.2%)	87 (17.4%)	29 (5.8%)

Table 2: Instance-level complementarity decomposition. "CoT only" means CoT is correct and A* is wrong; "A* only" means the reverse.

Instance-level decomposition. Table 2 makes the complementarity concrete. HotpotQA is especially striking: only 5.8% of instances are missed by both strategies, while 34.6% lie in a regime where exactly one strategy is correct. StrategyQA also exhibits sizable disagreement. LogicQA is harder overall, as shown by the 40.7% "both wrong" mass, but even there each strategy recovers failures of the other.

Scaling to 14B and math domains. To test generalization, we evaluate Qwen-2.5-14B on matched overlapping examples and LLaMA-3.1-8B on GSM8K and MATH (full results in Appendix G). At 14B scale, A* gives a large gain on HotpotQA (+23.14 pp) but slightly trails CoT on StrategyQA (−0.46 pp) and is roughly tied on LogicQA (+0.21 pp). Note that absolute CoT accuracy at 14B differs from 8B because the 14B evaluation uses only the overlapping subset where both models produced valid outputs (243 HotpotQA instances vs. the full 500), so cross-scale absolute comparisons should not be drawn. On math benchmarks, A* gives modest gains (+1.08 pp on GSM8K, +1.10 pp on MATH), smaller than on topology-heavy multi-hop QA. Both results confirm that the topology-dependent pattern persists regardless of model size or domain.

Cluster	Profile	BC	A*	CoT	Interpretation
C1	Linear, short	1.28	81.7	82.1	CoT sufficient
C2	Mid-branch	4.24	73.5	66.2	selection critical
C3	Complex, noisy	4.02	52.3	49.1	both often fail
C4	Deep-branch	6.49	69.8	58.4	A* largest gain

Table 3: Reasoning topology clusters. BC = branching coefficient. A* advantage grows monotonically with branching complexity.

3.2 Qualitative Analysis

Manual inspection of disagreement cases reveals recurring regimes (detailed examples in Appendix B). **A* helps** when reasoning has branching structure: (i) non-linear evidence aggregation, (ii) hypothesis elimination, and (iii) multi-hop composition—all cases where *deferred commitment* is valuable. **CoT remains preferable** on single-path reasoning: direct factual recall and coherent argumentation where branching fragments the reasoning frame. StrategyQA/HotpotQA contain many multi-hop instances (A* gains), while LogicQA requires stable logical frames (branching interferes). The tradeoff is best understood as *reasoning topology*: A* helps when the solution requires exploring multiple paths.

3.3 Explaining Complementarity Through Topology

We formalize the qualitative intuition using the topology features from Section 2.5. Applying k-means (k=4) to the 13 features across all instances yields four clusters (Table 3). A* advantage increases from −0.4 pp (C1) to +11.4 pp (C4), with $R^2 = 0.94$. Each cluster contains instances from multiple benchmarks, indicating the CoT/A* boundary is explained by structure, not dataset identity.

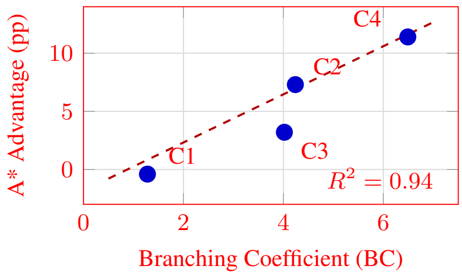

Figure 2: A* advantage (pp) vs. branching coefficient across topology clusters. Linear fit: $R^2 = 0.94$.

Router	Input	Str.QA	Log.QA	Hot.QA	Gap%
A* only	–	74.80	44.85	80.10	–
CoT only	–	64.52	45.78	76.80	–
Sem. Entropy	traces	74.37	47.47	81.30	14.8
LLM-as-Critic	traces	74.19	47.16	80.21	12.4
Random Forest	features	76.25	48.20	81.25	18.2
Topology Router	**13 feat.**	**77.35**	**49.01**	**82.41**	**26.1**
Oracle	–	84.59	59.29	94.20	100

Table 4: Router comparison with oracle gap recovery (%, averaged across datasets). The topology router observes no generated traces yet outperforms output-based routers on all three benchmarks.

Prescriptive decision rule. The cluster analysis yields a practical decision rule: *C1 (linear)*: use CoT; search overhead is unnecessary. *C2 (mid-branch)*: route adaptively; this is where router quality matters most. *C3 (complex, noisy)*: neither method is reliable; retrieval augmentation or stronger models may be needed. *C4 (deep-branch)*: use A*; search has the clearest advantage.

Topic modelling validates topology. As independent validation, we apply Non-negative Matrix Factorisation (NMF, k=8) on TF-IDF representations of all 3,660 paired evaluation questions. A* dominates in 6 of 8 topic categories covering 66.2% of the corpus—these share multi-hop entity chains (entertainment, historical events, geography, biography, politics). CoT-dominant topics (33.8%) correspond to single-step commonsense categories (everyday reasoning, hypothetical scenarios). Full per-topic results with examples in Appendix H.

3.4 Topology-Aware Routing

Table 4 compares all routers and reports oracle gap recovery.

Metric	A*	CoT
Average fluency score	0.725	0.673
Cases where more fluent	62.0%	38.0%
More-fluent but wrong	38.0% of disagreements	

Table 5: Fluency-Correctness Asymmetry on disagreement cases. Trace fluency does not predict correctness.

Why pre-generation routing outperforms post-hoc routing. The topology router observes no generated traces, yet outperforms routers that inspect both full reasoning traces. This reverses the intuitive expectation that seeing full traces should be more informative. The bottleneck is often not judging which completed trace looks better, but recognizing before generation which computation the problem requires.

Fluency-Correctness Asymmetry. We identify a fundamental limitation of output-based routing: in 38% of disagreement cases, CoT produces the *more fluent* trace but the *wrong* answer (Table 5). A critic that rewards coherence can therefore prefer the wrong computation. This asymmetry explains why the topology router (which sidesteps trace evaluation entirely) outperforms the LLM-as-critic.

3.5 Sensitivity, Compute, and Failure Analysis

Sensitivity analysis. On a stratified 300-question HotpotQA sample (seed=42), we sweep the node budget $B \in \{5, 10, 15, 20, 30\}$ and branching factor $K \in \{1, 2, 3, 4\}$ (full tables in Appendix E). Accuracy improves from B=5 (73.0%) to B=20 (78.7%), then drops at B=30 (76.0%), indicating diminishing returns. The default B=15 (78.0%) lies within 0.7 pp of the peak. Even K=1 (no real branching) achieves 77.7%, confirming graceful degradation and exceeding self-consistency.

Compute–accuracy tradeoff. A* at the default (K=3, B=15) uses ∼12,452 measured tokens/question (∼24× CoT's ∼512). A budget-reduced variant (K=1) achieves competitive accuracy at 3,372 tokens/question (∼7× CoT). A* is substantially more expensive; its use is justified only when topology predicts branching is useful, motivating pre-routing as a compute-saving mechanism (full table in Appendix F).

Failure analysis. Among A* failures on instances CoT solves correctly, 91.1% are *overthinking* (search continues past correct evidence, accumulating distractors) and 8.9% are *premature commitment* (search locks onto an early branch). CoT fails by committing too early; A* fails by continuing too long. The same branching mechanism that helps on multi-hop problems hurts on linear ones (examples in Appendix B.3).

Human evaluation. Two annotators labeled 100 disagreement instances per dataset on *coverage* (1–3) and *relevance* (1–3), with Cohen's κ: 0.59–0.68. The correct method consistently achieves higher coverage (e.g., StrategyQA: CoT-correct 3.00 vs. A* 1.20; A*-correct 2.71 vs. CoT 1.63), confirming that disagreements are driven by coverage of the required reasoning path rather than surface quality (full results in Appendix C).

4 Discussion

The CoT/A* boundary is structural, not dataset-specific. The strongest empirical finding is that A* advantage tracks branching coefficient across clusters (-0.4 pp in C1 to $+11.4$ pp in C4, Table 3), not benchmark labels. This means the question "should I use search?" is better answered by looking at the input's structure than by knowing which dataset it comes from.

Pre-generation routing beats trace inspection. This was the most surprising result to us. The topology router sees zero generated tokens yet outperforms critics that read both full traces. The explanation is the Fluency-Correctness Asymmetry: in 38% of disagreements, the wrong answer comes wrapped in the more fluent trace, fooling any judge that conflates readability with correctness. The implication is that adaptive inference should start with structural triage, not trace evaluation. The critic-based heuristic itself operates conservatively (mean 0.43 vs. human 0.58, $\epsilon_{\max} = 0.14$), providing practical confidence without formal admissibility.

Substantial room remains. The topology router recovers only 26.1% of the oracle gap. The remaining 74% likely involves factors we do not yet model: whether the base LLM has the right parametric knowledge for a given question, whether the search depth is matched to the problem, and how model capacity interacts with topology (the 14B results in Table 11 hint at complex interactions).

5 Related Work

CoT and structured reasoning. CoT (Wei et al., 2022), self-consistency (Wang et al., 2023), Tree-of-Thoughts (Yao et al., 2024), and Graph-of-Thoughts (Besta et al., 2024) explore increasingly rich reasoning structures. Recent work shows CoT is not universally beneficial (Meincke et al., 2025); we provide the search-side complement, identifying *when* search helps.

Search-augmented reasoning. MCTS-based reasoning (Hao et al., 2023), beam search (Xie et al., 2024), LATS (Zhou et al., 2024), MCTSr (Zhang et al., 2024), and rStar-Math (Guan et al., 2025) report aggregate search gains. We diagnose when gains occur and show they are topology-dependent.

Over-exploration and search failures. The overthinking survey (Sui et al., 2025), Fetch (Wang et al., 2025), and CoT-Valve (Ma et al., 2025) treat over-exploration as an efficiency issue. We show it is also a correctness issue: 91.1% of A* failures are overthinking failures where search actively degrades the answer.

Routing and critic limitations. Semantic entropy (Kossen et al., 2024) and LLM-as-judge methods (Zheng et al., 2023) route or evaluate post-hoc. Process supervision (Lightman et al., 2023) assumes trace quality is assessable from content. Mixture-of-experts (Shazeer et al., 2017) routes between neural modules. We route between *reasoning algorithms* based on input structure, and our Fluency-Correctness Asymmetry finding (38% of fluent traces are wrong) establishes a ceiling on output-based routing. Unlike prior methods, our contribution is a diagnostic framework for understanding *when* search helps.

6 Conclusion

Across five benchmarks and three model scales, we find that CoT and A* are complementary rather than hierarchical: the gap between the oracle and the best single method is consistently large. The key explanatory variable is branching coefficient, not dataset identity—a structural finding that holds across both QA and math domains. Perhaps most surprisingly, a router that sees only the raw input (no traces) outperforms critics that inspect both full reasoning chains, because trace fluency misleads post-hoc judges in a systematic way. These results suggest that the next step for adaptive inference is not better trace evaluation but better problem-structure recognition. Concrete next steps include learned topology embeddings and extending the analysis to code generation.

Limitations

Our findings are conditioned on a single budgeted A* implementation; a different search policy (MCTS, beam search) could shift where the complementarity boundary lies. The 13 topology features are hand-designed proxies—interpretable but unlikely to capture the full latent structure of reasoning difficulty. They also rely on English-language parsing tools, limiting direct transfer to other languages or code. The critic-based heuristic depends on DeepSeek-R1-Distill-Qwen-14B; although the topology router reduces this dependence, it does not eliminate it entirely. On the cost side, A* at default settings uses $\sim 24\times$ the tokens of CoT, and our token accounting covers only a 300-question ablation sample. Domain coverage now includes math (GSM8K, MATH) but coding and long-form generation remain untested. Finally, the topology router is an XGBoost classifier on fixed features; a learned router that jointly optimizes routing and search would be a natural next step.

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A Auxiliary Conservative Heuristic Construction

This appendix presents a formal admissibility proof for an auxiliary depth-plus-coverage heuristic. This construction applies to a simplified setting and is *not* used as the main experimental heuristic (which is the critic-based heuristic described in Section 2.4). We include it for theoretical completeness.

The auxiliary heuristic combines a depth-based term with a coverage term:

$$h_{\text{aux}}(n) = \omega_d \max(0, H_{\min} - d_n) + \omega_c(1 - E(n)), \tag{8}$$

where $H_{\min}$ is a task-level lower bound on reasoning hops, $E(n) \in [0,1]$ is the fraction of target entities covered, and $\omega_d, \omega_c > 0$ are weights. Let $c_{\min} = 1 + (1 - c_{\max\_\text{conf}})$ denote the minimum transition cost.

Assumption 1 (Minimum-hop lower bound). *Any valid goal trace requires at least $H_{\min}$ reasoning steps from the root.*

Assumption 2 (Non-goal continuation). *Any non-goal node with depth $d_n \geq H_{\min}$ still requires at least one additional transition.*

Theorem 1 (Conservative admissibility). *Under Assumptions 1 and 2, if $\omega_d + \omega_c \leq c_{\min}$, then $h_{aux}(n) \leq h^*(n)$ for all non-goal nodes.*

Proof. Case 1 (root, $d_n = 0$): $h^*(n) \geq H_{\min} c_{\min}$. Since $h_{\text{aux}}(n) \leq \omega_d H_{\min} + \omega_c \leq H_{\min} c_{\min}$. Case 2 ($0 < d_n < H_{\min}$): Let $m = H_{\min} - d_n$. Then $h^*(n) \geq m c_{\min}$ and $h_{\text{aux}}(n) \leq \omega_d m + \omega_c \leq m c_{\min}$. Case 3 ($d_n \geq H_{\min}$): $h^*(n) \geq c_{\min}$ and $h_{\text{aux}}(n) = \omega_c(1 - E(n)) \leq \omega_c \leq c_{\min}$. □

B Qualitative Examples

This appendix provides the detailed examples corresponding to the qualitative conclusions summarized in Section 3.2.

B.1 Patterns of A* Success and CoT Failure

We identify three recurring settings where A* succeeds while CoT fails.

Group 1: Non-linear evidence aggregation. These are questions where the answer requires combining several facts that do not naturally form a single left-to-right narrative.

StrategyQA — Non-linear Evidence Aggregation

Q: *Is shrimp scampi definitely free of plastic?* (Gold: NO)

CoT (× predicts YES): Proceeds through five sequential steps analyzing harvesting, processing, cooking, ingredients, and regulations. Each step independently concludes no significant plastic risk. The linear struc-

Hyperparameter	Symbol	StrategyQA	LogicQA	HotpotQA
Heuristic form	Eq. (6)–(8)	$h_{\text{depth}} + h_{\text{coverage}}$	$\frac{1}{2}(h_{\text{remaining}} + h_{\text{logic}})$	$h_{\text{depth}} + h_{\text{coverage}}$
Minimum hop lower bound	$H_{\min}$	2	–	2
Depth weight	ω_d	0.3	–	0.3
Coverage weight	ω_c	0.3	–	0.3
Logic-complexity weight	λ	–	0.12	–
Confidence cap	$c_{\max\_\text{conf}}$	0.99	0.99	0.99
Branching factor	N_c	2	2 (`n_candidates`)	3
Maximum search depth	$D_{\max}$	8	5	8
Maximum explored nodes	B_{nodes}	15	12	30
Minimum goal depth	–	2	–	2
Vote threshold	–	2 matching answers	2 goal nodes	2 matching answers
Branch temperatures	–	`[0.2, 0.6]`	0.7 (normal), 0.05 (forced)	`[0.15, 0.45, 0.75]`
Forced conclusion trigger	–	$d \geq D_{\max} - 1$	$d \geq 4$	$d \geq D_{\max} - 1$
Early goal guard	–	if $d < 2$: conf. ≥ 0.75	–	if $d < 2$: conf. ≥ 0.85, grounding ≥ 0.5
Ungrounded-answer penalty	–	–	–	+0.8 path cost if grounding < 0.3

Table 6: A* search hyperparameters extracted from the dataset-specific implementations. StrategyQA and HotpotQA use the depth-plus-coverage heuristic from Section 2.4, whereas LogicQA uses a distinct heuristic based on remaining depth and logical complexity.

ture prevents cross-referencing marine biology evidence with the food preparation chain.

A* (✓ predicts NO): Step 1 directly identifies microplastic bioaccumulation in shrimp tissue. Step 2 notes that cooking does not remove microplastics. The heuristic guides the search toward the most relevant evidence node, avoiding the exhaustive but misleading step-by-step analysis.

LogicQA — Non-linear Evidence Aggregation

Q: *Based on the above statement [spatial arrangement of communities around Civic Park], which of the following can be derived?* (Gold: A)

CoT (× predicts D): Analyzes spatial relationships sequentially, accumulating errors in relative positioning. By the fourth step, the accumulated imprecision leads to selecting an incorrect spatial relationship.

A* (✓ predicts A): Explores multiple spatial constraint combinations in parallel. The search finds a consistent assignment by simultaneously satisfying the "southwest of" and "southeast of" constraints, correctly deriving that Civic Park is north of the administrative service area.

Group 2: Hypothesis elimination. A* is also effective when the problem naturally decomposes into evaluating or eliminating alternatives.

LogicQA — Hypothesis Elimination

Q: *Which of the following is most similar to the argument: "All landscape rooms can see the landscape, but Li Wenbing's family can't see the landscape. Therefore, Li Wenbing's family is not a landscape room."* (Gold: D)

CoT (× predicts C): Identifies the logical structure (modus tollens) but then incorrectly maps it to option C (a modus ponens structure) due to surface similarity.

A* (✓ predicts D): Separately evaluates options through different branches. One branch identifies the negative inference pattern in the premise and correctly maps it to option D ("Anyone who meets conditions can apply; Sun Wen did not apply; therefore Sun Wen did not meet conditions"), which preserves the modus tollens structure.

Group 3: Multi-hop question answering. HotpotQA highlights the value of search when the answer depends on chaining facts across sources.

HotpotQA — Multi-hop Chaining

Q: *What government position was held by the woman who portrayed Corliss Archer in the film Kiss and Tell?* (Gold: Chief of Protocol)

CoT (× predicts "Ambassador"): Correctly identifies Shirley Temple as the actress but then retrieves an incorrect government position from parametric memory, confusing ambassadorial roles with her protocol role.

A*: Explores multiple evidence paths for Shirley Temple's government career. While the trace is imperfect, the tree search structure allows it to consider multiple candidate positions before committing.

B.2 Patterns of CoT Success and A* Failure

We identify two common settings where CoT remains better than search.

Group 1: Direct factual reasoning. When the answer follows directly from familiar facts, search can introduce unnecessary branching and semantic drift.

StrategyQA — Direct Factual Reasoning

Q: *Is a Boeing 737 cost covered by Wonder Woman (2017 film) box office receipts?* (Gold: Yes)

CoT (✓ predicts Yes): Directly retrieves the production budget (∼$150M), box office (∼$822M), and Boeing 737 cost (∼$100M), concluding that the receipts exceed the aircraft cost.

A* (× predicts No): The first search step focuses on whether box office receipts are "associated with" aircraft purchases, misreading the question as one about financial transactions rather than magnitude comparison.

StrategyQA — Direct Factual Reasoning

Q: *Is average number of peas in a pod enough commas for a billion?* (Gold: Yes)

CoT (✓ predicts Yes): Correctly identifies that a billion (1,000,000,000) requires three commas and that a pea pod usually contains 7–9 peas, so the answer is yes.

A* (× predicts No): Incorrectly states that a billion has 12 zeros and becomes confused by the scale comparison.

Group 2: Fluent world knowledge and argumentation. When the problem is best solved by a smooth, coherent narrative, CoT often remains more reliable than fragmented search steps.

LogicQA — Fluent Argumentation

Q: *Which of the following is the best argument against the claim that rich Chinese are transferring property abroad?* (Gold: B)

CoT (✓ predicts B): Systematically compares the answer options and identifies the strongest counterargument.

A* (× predicts A): The first search step identifies option A as relevant, and the heuristic keeps the search near that branch without sufficiently comparing the full option set.

B.3 A* Failure Modes on Simple Cases

We observe three recurrent failure modes for A* on examples that CoT solves easily.

Failure mode 1: Overthinking simple questions.

StrategyQA — Overthinking

Q: *Does Disney have an ice princess?* (Gold: YES)

A* (× predicts NO): Step 1 correctly identifies Elsa from *Frozen* as having ice powers. Step 2 notes that she is called "The Ice Queen" in promotional material. The search then turns the easy recall problem into an unnecessary semantic distinction between "queen" and "princess," yielding the wrong answer.

Failure mode 2: Spurious correlations in search.

StrategyQA — Spurious Correlation

Q: *Is Menthol associated with Thanksgiving?* (Gold: NO)

A* (× predicts YES): Step 1 associates menthol with cough drops used in cold weather. Step 2 associates Thanksgiving with cold weather in North America. Step 3 turns these weak associations into a spurious chain, effectively reasoning menthol → cold weather → Thanksgiving.

Failure mode 3: Step repetition.

StrategyQA — Step Repetition

Q: *Will the Albany in Georgia reach a hundred thousand occupants before the one in New York?* (Gold: NO)

A* (× predicts YES): Step 2 states the population of Albany, New York (∼98,000). Step 3 largely repeats the same fact. The repetition wastes node budget that should have been used to compare growth trajectories.

C Human Evaluation Details

Two annotators labeled 100 disagreement instances per dataset on *coverage* (1–3: does the trace reach the required inferential path?) and *relevance* (1–3: stays on-topic?). Cohen's κ: 0.59–0.68.

Algorithm 1 Budgeted A* Search over Reasoning Traces

Require: Problem $\mathcal{Q} = (q, C, \mathcal{O})$, model $\mathcal{M}_\theta$, depth limit $D_{\max}$, node budget B, candidates N_c

Initialize root $n_0 \leftarrow (\emptyset, 0)$, $g(n_0) \leftarrow 0$, $h(n_0) \leftarrow h(n_0)$ from Eq. (6)
OPEN $\leftarrow \{n_0\}$ (priority queue on $f(n) = g(n) + h(n)$)
VISITED $\leftarrow \emptyset$, GOALS $\leftarrow \emptyset$, explored $\leftarrow 0$
while OPEN $\neq \emptyset$ and explored $< B$ **do**
  $n \leftarrow$ OPEN.pop_min()
  if sig$(n) \in$ VISITED **then**
    continue
  end if
  VISITED $\leftarrow$ VISITED $\cup \{\text{sig}(n)\}$
  explored $\leftarrow$ explored $+1$
  if ISGOAL(n) **then**
    GOALS $\leftarrow$ GOALS $\cup \{n\}$
    continue
  end if
  if $d_n \geq D_{\max}$ **then**
    continue
  end if
  $\{(s^{(j)}, c^{(j)})\}_{j=1}^{N_c} \leftarrow$ GENERATESTEPS$(\mathcal{M}_\theta, q, C, \mathcal{O}, \tau_n, N_c)$
  for each $(s^{(j)}, c^{(j)})$ **do**
    $n' \leftarrow (\tau_n \oplus s^{(j)}, d_n + 1)$
    $g(n') \leftarrow g(n) + 1 + (1 - c^{(j)})$
    OPEN.push$(n', g(n') + h(n'))$
  end for
end while
return weighted vote over GOALS

The correct method consistently receives higher coverage. Relevance is weakly diagnostic: a trace can stay on-topic yet miss decisive steps.

D FAQs

Q1. Why are gains on HotpotQA larger than StrategyQA? HotpotQA explicitly rewards multi-hop evidence aggregation; StrategyQA is often solvable linearly.

Q2. Why include LogicQA where A* underperforms? LogicQA stresses formal sequential reasoning where branching can interfere, giving a more honest picture.

Q3. How reliable are self-reported confidence scores? They are pragmatic ranking signals (not

LLM-as-Judge

Purpose. Compare two candidate reasoning trajectories and decide which one solves the query more *faithfully* and *correctly*.

Inputs.

- Context
- Query
- Path A reasoning trace
- Path B reasoning trace

Evaluation criteria.

- **Faithfulness:** uses only the provided context and avoids invented constraints
- **Logical validity:** each step follows causally and coherently from prior steps
- **Grounded conclusion:** the final answer directly resolves the query from the established reasoning

Required output. The model first reasons internally about both traces, then returns only the winning path in constrained XML:

`<winner>A</winner>` or `<winner>B</winner>`

LLM-as-Critic

Purpose. Diagnose the weaknesses of two reasoning traces and determine which trace is better overall.

Inputs.

- Question
- Answer options
- Trace A with predicted answer
- Trace B with predicted answer

Evaluation criteria.

- whether the reasoning actually entails the predicted answer
- whether there are logical fallacies or unsupported jumps
- whether any claims are hallucinated or ungrounded

Required output. The model returns a structured JSON critique with flaws for both traces and the preferred trace:

```
{"trace_A_flaws": "...",
 "trace_B_flaws": "...",
 "better_trace": "A/B"}
```

Figure 3: LLM-based routing prompts. The *judge* prompt is verdict-oriented: it compares two reasoning paths under faithfulness and logical soundness criteria and outputs only the winner. The *critic* prompt is diagnosis-oriented: it identifies flaws in each trace before selecting the better one in structured JSON form.

	Group	n	CoT Cov.	A* Cov.
Str.	CoT right	51	3.00	1.20
	A* right	38	1.63	2.71
Log.	CoT right	36	3.00	1.86
	A* right	26	1.88	2.77

Table 7: Human annotation (coverage, 1–3 scale) on sampled disagreement cases. The correct method consistently achieves higher reasoning coverage.

calibrated probabilities) biasing search toward shorter, more confident traces.

Q4. How should readers interpret the admissibility result? The formal proof (Appendix A) applies only to the auxiliary depth-coverage heuristic under explicit assumptions. The main experimental heuristic is empirically validated (Section 2.4) but not formally admissible.

Q5. Would ensemble voting capture the same complementarity? Only partially. Self-consistency recovers diversity but does not guide exploration toward promising reasoning states.

Q6. Is goal detection based on answer patterns brittle? Somewhat. Pattern matching is fast and reproducible but can miss valid conclusions phrased differently.

E Sensitivity Analysis

All experiments on a fixed stratified 300-question HotpotQA sample (seed=42).

Budget B	5	10	15	20	30
Accuracy (%)	73.0	77.3	78.0	78.7	76.0

Table 8: Budget sweep. Clear monotonic improvement B=5→20 (+5.7 pp), diminishing returns at B=30.

Branching K	1	2	3	4
Accuracy (%)	77.7	78.3	76.0	80.3

Table 9: Branching factor sweep. K=1 (linear chain) degrades gracefully; K=4 achieves the highest accuracy.

Method	Acc.%	Tok/q	Notes
CoT	76.80	∼512	single call
SC (5 traces)	78.21	∼2,560	5× CoT
A*, K=1, B=30	77.7	3,372	linear chain
A*, K=3, B=15	78.0	12,452	paper default
A*, K=3, B=20	78.7	12,391	peak accuracy

Table 10: Compute–accuracy tradeoff. Token counts are measured from ablation experiments, not estimated.

Setting	Dataset	CoT	A*	Δ
14B (Qwen-2.5-14B)	StrategyQA	76.02	75.56	−0.46
	HotpotQA	60.19	83.33	+23.14
	LogicQA	57.91	58.12	+0.21
Math (LLaMA-3.1-8B)	GSM8K	85.04	86.12	+1.08
	MATH	78.00	79.10	+1.10

Table 11: Additional results: 14B-scale on matched overlapping examples and math reasoning benchmarks. A* gains are topology-dependent across both scale and domain.

F Compute Analysis

G Additional Model and Domain Results

H Cross-Dataset Topic Modelling

We apply Non-negative Matrix Factorisation (NMF; Lee and Seung, 1999) to all 3,660 paired question texts from the 14B Qwen-2.5 evaluation (StrategyQA 2,290; HotpotQA 719; LogicQA 651).

Pipeline. (1) Text extraction: question string for StrategyQA/HotpotQA, scenario context for LogicQA (since its query strings are formulaic). (2) TF-IDF: max_df=0.80, min_df=3, max_features=5,000, ngram_range=(1,2), English stop-words removed. (3) NMF: k=8, init=NNDSVD-a, max_iter=500, random_state=42. (4) Topic assignment via argmax. (5) Per-topic A*/CoT accuracy.

Validation. We select k=8 via NPMI coherence (0.301), the smallest k above 0.30 with interpretable topics.

Key finding. A* wins in 6 of 8 categories covering 66.2% of the corpus (Table 12). All A*-dominant topics share a structural property: they require resolving multi-hop entity or factual chains. The two CoT-dominant categories—everyday_reasoning and hypothetical_scenario—involve commonsense judgements where a single inference step suffices.

Topic	Cov.%	Str.QA (A*/CoT)	Hot.QA (A*/CoT)	W
everyday_reasoning	27.7	79/79	88/73	CoT
entertainment_media	23.9	73/73	83/59	A*
historical_event	12.7	78/78	68/47	A*
geography_location	12.6	70/74	86/55	A*
general_property	9.9	73/72	63/40	A*
hypothetical_scenario	6.1	72/76	100/57	CoT
us_politics_law	4.4	72/81	91/36	A*
biography_origin	2.7	65/70	90/63	A*

Table 12: Per-topic A* vs. CoT accuracy (%) on 14B Qwen-2.5. W = overall winner across datasets.

entertainment_media (A* +10.4 pp): "If you were on a diet, would you have to skip lunch at McDonald's?" requires resolving menu composition plus dietary constraints across multiple facts.

historical_event (A* +2.4 pp): "Did Harry Houdini's wife make psychics look foolish?" requires chaining Houdini's death → wife's debunking campaign.

geography_location (A* +9.5 pp): "Is Disney associated with Los Angeles County?" requires resolving corporate headquarters location through entity disambiguation.

everyday_reasoning (CoT +0.5 pp): "Is capturing giant squid in natural habitat impossible with no gear?" is answerable in a single commonsense step.

hypothetical_scenario (CoT +2.7 pp): "Were deaths from Apollo 13 eclipsed by other space missions?" requires a single factual comparison.

I Reproducibility Details

Models. LLaMA-3.1-8B-Instruct (main backbone), Qwen2-0.5B-Instruct (small-scale comparison), Qwen-2.5-14B-Instruct (scale experiment), DeepSeek-R1-Distill-Qwen-14B (critic/heuristic). All models served locally via Ollama (v0.3.x) or vLLM (v0.4.x) with greedy decoding for CoT and temperature sampling for A* branches (per-dataset temperatures in Table 6).

Datasets. StrategyQA: full dev set, 2,290 examples. LogicQA: full test set, 651 examples. HotpotQA: stratified random sample of 500 from the fullwiki dev set (seed=42). GSM8K: test split, 1,319 examples. MATH: test split, 5,000 examples. 14B experiments: overlapping subset where both 8B and 14B produced parseable outputs (243–651 per dataset).

Hardware. All experiments run on a single node with 2× NVIDIA A100 80GB GPUs. Total compute for full experimental pipeline: approximately 340 GPU-hours.

Seeds and variance. CoT uses greedy decoding (deterministic). A* uses temperature-based branching; we report single-run results but the sensitivity analysis (Appendix E) covers hyperparameter variance. Router training uses 5-fold cross-validation with stratified splits (seed=42).

Code. Full pipeline code (data preprocessing, A* search implementation, router training, evaluation scripts) is available at the anonymous repository linked in the abstract.